

Unit: I Linear Programming.

• Introduction :-

Linear Programming problem (LPP) is a mathematical technique used to make the best possible decision under given limitations.

In Simple words,

- i) When resources are limited (time, money, machines, labour) and we want to maximize profit or minimize cost, we use linear programming.

For Example,

Suppose you have, limited raw materials, limited workers, limited time and you want to earn maximum profit. Linear Programming helps you decide how to use these resources efficiently.

• Applications of Linear Programming :-

1) Product mix Problem :-

Deciding how much quantity of each product should be produced to get maximum profit.

2) Diet Planning Problem :-

Preparing a healthy diet at minimum cost.

3) Cargo loading Problem :-

Loading goods in a truck or ship in the best possible way considering weight and space.

4) Capital Budgeting Problem :-

Allocating funds to different projects to get maximum return.

5) Manpower Planning Problem :-

Assigning workers efficiently to different jobs.

• Concept of linear Programming Model :-

Every linear programming problem consist of the following Components:

1) Decision Variables

2) Objective function

3) Constraints

4) Non-Negative Restrictions.

1) Decision Variables :-

Decision Variables represent the Unknown quantities that we need to determine.

Example: a) x_1 = number of chairs produced

b) x_2 = number of tables produced.

2) Objective function :-

The objective function represents what we want to:

• Maximize (profit, output)

• Minimize (Cost, time)

For Example;

Objective function is,

$$\text{Maximize } Z = 50x + 40y.$$

3) Constraints :-

Constraints are the limitations or restrictions on resources.

For Example:

constraint equation,

$$2x + y \leq 100$$

• Final Linear Programming Model:-

$$\text{Maximize } Z = 250x_1 + 400x_2$$

Subject to,

$$2x_1 + 5x_2 \leq 200$$

$$4x_1 + 2x_2 \leq 240$$

$$x_1 \geq 0, x_2 \geq 0$$

* Properties of Linear Programming Solution :-

1) Feasible Solution :-

A feasible solution is any solution that satisfies all constraints and non-negativity conditions.

Simply, A solution that is allowed and possible.

2) Optimal Solution :-

An optimal solution is the best feasible solution which gives:

- Maximum value (in maximization problems)

- Minimum value (in minimization problems)

Simply, The solution that gives maximum profit or minimum cost.

3) Alternate Optimal Solutions:-

When more than one feasible solution gives the same maximum (or minimum value), they are called alternate optimal solutions.

Simply, Different solutions, same best profit.

4) Unbounded Solution :-

For some LPP, the objective function value can be increased /

Decreased infinitely without any limitation. Such solution is known as unbounded solution.

Example, profit keeps increasing no upper limit.

5) Infeasible Solution :-

A problem is infeasible if no solution satisfies all constraints at the same time.

Simply, No possible solution.

6) Degenerate Solution :-

A degenerate solution is a solution in which one or more basic variables have value zero.

* Development of LP Models :-

In Linear programming, modelling means formulating an objective function and constraints.

Example :- Formulate the linear programming model for following situation.

A manufacture produces two products A and B. Each unit of product A requires 2 hours of machine A time and 1 hour of labour time. Each unit of product B requires 1 hour of machine time and 2 hours of labour time.

The total available machine time is 100 hours and total available labor time is 80 hours. Profit per unit of product A is 50rs and product B is 40rs.

Soln:- Let, the Decision variables are,

x_1 = Number of units of product A.

x_2 = Number of units of product B.

Now, the objective function is, to maximize total profit,

$\therefore$ Maximize $Z = 50x_1 + 40x_2$ — ①

Constraints:-

II) Simplex Method :-

Example:- Solve by using Simplex Method.

$$\text{Maximize, } Z = 6x_1 + 8x_2$$

$$\text{Subject to, } 5x_1 + 10x_2 \leq 60$$

$$4x_1 + 4x_2 \leq 40$$

$$x_1, x_2 \geq 0$$

Soln:- The standard form of above LP model is,

$$\text{Maximize, } Z = 6x_1 + 8x_2 + 0x_3 + 0x_4$$

Subject to,

$$5x_1 + 10x_2 + x_3 + 0x_4 = 60$$

$$4x_1 + 4x_2 + 0x_3 + x_4 = 40$$

$$x_1, \dots, x_4 \geq 0$$

where, x_3 and x_4 are slack variable.
and x_3 and x_4 are basic variables.

- Canonical form:- It is the form in which each constraint has basic variable.

- Basic Variable :-

A variable is said to be basic variable if it has unit coefficient in one of the constraints and zero coefficient in the remaining constraints.

Step : 2

Table : 1

Initial Simplex Table:-

CB_i	C_j	6	8	0	0	Solution	Ratio
	Basic variable	x_1	x_2	x_3	x_4		
0	$\vec{x_3}$	5	10	1	0	60	$\frac{60}{10} = 6$
0	x_4	4	4	0	1	40	$\frac{40}{4} = 10$
	Z_j	0	0	0	0		
	$C_j - Z_j$	6	8 ↑	0	0		

Where, C_j is the coefficient of the j th term of obj. function.
 CB_i is the coefficient of i th basic variable in the obj. fn.

$$\text{Now, } Z_j = \sum_{i=1}^2 (CB_i)(a_{ij})$$

$$\therefore Z_1 = CB_1 \cdot a_{11} + CB_2 \cdot a_{21}$$

$$= 0 \times 5 + 0 \times 4$$

$$\boxed{Z_1 = 0}$$

$$\text{and, } Z_2 = CB_1 \cdot a_{12} + CB_2 \cdot a_{22}$$

$$= 0 \times 4 + 0 \times 4$$

$$\boxed{Z_2 = 0}$$

- Optimality Condition :- 1) For maximization Problem, if all $C_j - Z_j$ are less than or equal to zero, then optimality is reached. Otherwise select the variable with maximum $C_j - Z_j$ value as the entering variable.

- 2) For minimization Problem, if all $C_j - Z_j$ are either equal to or greater than zero, the optimality is reached; otherwise select the variable with the most negative value as the entering variable.

In above maximization problem, all $C_j - Z_j$ are greater equal to zero and x_2 is maximum $C_j - Z_j$ value is maximum $\therefore x_2$ is entering variable.

$\therefore x_2$ enters the basis and corresponding column is called as key column.

• Feasibility Condition :-

1) In each row, find the ratio between the solution column value and the value in the key column.

2) Then select the variable from the present set of basic variables with respect to the minimum ratio.

Such a variable is leaving variable and corresponding row is called as the key row.

• key element or Pivot element :-

The value at the intersection of key row and key column is called as key element or pivot element.

In above problem, x_3 is leaving variable as having minimum ratio, $\therefore x_3$ is leaving variable.

$\therefore$ Row 1 is the key row.

and key element or pivot element is 10.

Step: III) $\mathbb{I}^{\text{st}}$ Simplex Table

	C_j	6	8	0	0		
C_B	Basic variable	x_1	x_2	x_3	x_4	Solution	Ratio
8	x_2	$\frac{1}{2}$	1	$\frac{1}{10}$	0	6	$\frac{6}{(1/2)} = 12$
0	x_1	2	0	$-\frac{2}{5}$	1	16	$16/2 = 8$
	Z_j	4	8	$\frac{4}{5}$	0	48	
	$C_j - Z_j$	2	0	$-\frac{4}{5}$	0		

Here, we perform operation on $R_1 \rightarrow \frac{R_1}{2}$

and on $R_2 \rightarrow R_2 - 4R_1$

Step: IV) $\mathbb{II}^{\text{nd}}$ Simplex Table

	C_j	6	8	0	0		
C_B	Basic variable	x_1	x_2	x_3	x_4	Solution	
8	x_2	0	1	$\frac{1}{5}$	$-\frac{1}{4}$	2	
6	x_1	1	0	$-\frac{1}{5}$	$\frac{1}{2}$	8	
	Z_j	6	8	$\frac{2}{5}$	$-\frac{1}{2}$	64	
	$C_j - Z_j$	0	0	$-\frac{2}{5}$	$-\frac{1}{2}$		

In above table, all the values of $C_j - Z_j$ are either zero or negative.

Hence, the optimality is reached.

Hence, the soln is, $x_2 = 2, x_1 = 8$

$\therefore$ and max $Z = 64$

2) Solve the following LPP by using simplex Table.

$$\max Z = 10x_1 + 15x_2 + 20x_3$$

$$\text{s.t. } 2x_1 + 4x_2 + 6x_3 \leq 24$$

$$3x_1 + 9x_2 + 6x_3 \leq 30$$

$$x_1, x_2, x_3 \geq 0$$

Soln:- The standard form of above LPP is,

$$\max Z = 10x_1 + 15x_2 + 20x_3 + 0x_4 + 0x_5$$

s.t

$$2x_1 + 4x_2 + 6x_3 + x_4 + 0x_5 = 24$$

$$3x_1 + 9x_2 + 6x_3 + 0x_4 + x_5 = 30$$

$$x_1, x_2, \dots, x_5 \geq 0$$

Here, x_4, x_5 are slack variables.

Step: II Initial simplex Table

C _B i	C _j	10	15	20	0	0	soln	Ratio
	Basic variable	x_1	x_2	x_3	x_4	x_5		
0	$\xrightarrow{x_4}$	2	4	6	1	0	24	$\frac{24}{6} = 4$
0	x_5	3	9	6	0	1	30	$\frac{30}{6} = 5$
	Z_j	10	0	0	0	0	0	
	$C_j - Z_j$	10	15	20	0	0		

Here, Not all $C_j - Z_j \leq 0$.

Hence, above solution is not optimum.

Here, max positive is 20, $\therefore x_3$ enters the basis.

min ratio is 4, $\therefore x_4$ leaves the basis.

and 6 is a pivot element

Step: III) Ist Simplex Table

CB _i	C _j	10	15	20	0	0	Soln	Ratio
	x _B	x ₁	x ₂	x ₃	x ₄	x ₅		
20	x ₃	1/3	2/3	1	1/6	0	4	12
0	$\vec{x_5}$	1	5	0	-1	1	6	6
	Z _j	20/3	40/3	20	10/3	0		
	C _j - Z _j	10/3 ↑	5/3	0	-10/3	0		

Here, not all $C_j - Z_j \geq 0$.

∴ Above soln is not optimum.

x₅ leaves the basis.

x₁ enters the basis.

and 1 is a pivot element.

Step: IV) IInd Simplex Table

CB _i	C _j	10	15	20	0	0	Soln
	x _B	x ₁	x ₂	x ₃	x ₄	x ₅	
20	x ₃	0	-1	1	1/2	-1/3	2
10	x ₁	1	5	0	-1	1	6
	Z _j	10	30	20	0	10/3	100
	C _j - Z _j	0	-15	0	0	-10/3	

Here, all $C_j - Z_j \leq 0$. ∴ Above soln is optimum.

soln is, $x_1 = 6, x_2 = 2$ and $x_3 = 10$.

Ex.1) Solve the following LPP by using Simplex Method. (10:45:10)

$$\text{Minimize } Z = x_1 - 3x_2 + 2x_3$$

$$\text{s.t. } 3x_1 - x_2 + 2x_3 \leq 7$$

$$-2x_1 + 4x_2 \leq 12$$

$$-4x_1 + 3x_2 + 8x_3 \leq 10$$

$$x_1, x_2, x_3 \geq 0$$

Soln:- Write Step: I) We write above LPP in standard form.

So,

$$\text{minimize } Z = x_1 - 3x_2 + 2x_3$$

s.t.

$$3x_1 - x_2 + 2x_3 + x_4 + 0x_5 + 0x_6 = 7$$

$$-2x_1 + 4x_2 + 0x_3 + 0x_4 + x_5 + 0x_6 = 12$$

$$-4x_1 + 3x_2 + 8x_3 + 0x_4 + 0x_5 + x_6 = 10$$

$$x_1, \dots, x_6 \geq 0$$

where, x_4, x_5 and x_6 are slack variable and basic variable

Step: II) Initial Simplex Table

C _B	C _J	1	-3	2	0	0	0	Soln	Ratio
	Basic variable	x_1	x_2	x_3	x_4	x_5	x_6		
0	x_4	3	-1	2	1	0	0	7	
0	x_5	-2	4	0	0	1	0	12	$\frac{12}{4} = 3$
0	x_6	-4	3	8	0	0	1	10	$\frac{10}{3} = 3.33$
	Z	0	0	0	0	0	0	0	
	C _J - Z _j	1	-3	2	0	0	0	0	

Here x_2 enters the basis

and x_5 leaves the basis.

and 4 is the pivot element

Step: III) Ist Simplex Table

C _B i	C _j	1	-3	2	0	0	0	Sol ⁿ	Ratio
	Basic Variable	x ₁	x ₂	x ₃	x ₄	x ₅	x ₆		
0	$\vec{x_4}$	$\boxed{5/2}$	0	2	1	$1/4$	0	10	4
-3	x ₂	$-1/2$	1	0	0	$1/4$	0	3	—
0	x ₆	$-5/2$	0	8	0	$-3/4$	1	1	—
	Z _j	$3/2$	-3	0	0	$-3/4$	0	-9	
	Z _j -Z _i	$-1/2 \uparrow$	0	2	0	$3/4$	0		

Here, x₁ enters the basis.

x₄ leaves the basis.

and, $5/2$ pivot element.

Step: IV) IInd Simplex Table

C _B i	C _j	1	-3	2	0	0	0		
	Basic Variable	x ₁	x ₂	x ₃	x ₄	x ₅	x ₆	Sol ⁿ	Ratio
-1	x ₁	1	0	$4/5$	$2/5$	$1/10$	0	4	
3	x ₂	0	1	$2/5$	$1/5$	$1/2$	0	5	
0	x ₆	0	0	10	1	$-1/2$	1	11	
	Z _j	1	-3	$-2/5$	$-1/5$	$-7/5$	0	-11	
	Z _j -Z _i	0	0	$+12/5$	$+1/5$	$+7/5$	0		

$\therefore$ all $Z_j - Z_i \geq 0$, we get the optimum solⁿ and the solⁿ is

$$x_1 = 4, x_2 = 5, x_6 = 11$$

$$\text{and minimize } Z = 4 - 3(5) = -11.$$

* Unbounded Solution :-

Note:- 1) When value of objective function increases infinitely, we say that given problem has an unbounded soln.

2) In a simplex table if any column of decision variable has all the entries ≤ 0 then we say there exist an unbounded soln.

Que. 1) Solve the following LPP by Simplex Method.

$$\text{Max } Z = 2x_1 - x_2 + x_3$$

s.t.

$$x_1 + x_2 - 3x_3 \leq 8$$

$$-4x_1 + x_2 - x_3 \leq 2$$

$$-2x_1 - 3x_2 + x_3 \leq 4$$

$$x_1, x_2, x_3 \geq 0$$

Soln:- Converting the problem to standard form, we have,

$$\text{Max } Z = 2x_1 - x_2 + x_3 + 0x_4 + 0x_5 + 0x_6$$

s.t.

$$x_1 + x_2 - 3x_3 + x_4 + 0x_5 + 0x_6 = 8$$

$$-4x_1 + x_2 - x_3 + 0x_4 + x_5 + 0x_6 = 2$$

$$-2x_1 - 3x_2 + x_3 + 0x_4 + 0x_5 + x_6 = 4$$

$$x_1, x_2, \dots, x_6 \geq 0$$

Step: II) Initial Simplex Table

CB	Cj	2	-1	1	0	0	0		
	Basic Variable	x_1	x_2	x_3	x_4	x_5	x_6	soln	Ratio
0	$\overrightarrow{x_4}$	1	1	-3	1	0	0	8	8
0	x_5	-4	1	-1	0	1	0	2	
0	x_6	-2	-3	1	0	0	1	4	
	Z_j	0	0	0	0	0	0	0	
	$C_j - Z_j$	2	-1	1	0	0	0		

Step: III) 1st Simplex Table

CB	C _j Basic variable	2	-1	0	0	0	0	Soln
2	x ₁	1	1	-3	1	0	0	8
0	x ₅	0	5	-13	4	1	0	34
0	x ₆	0	-1	-5	2	0	1	20
	Z _j	2	-1	-6	2	0	0	16
	C _j - Z _j	0	0	5	-2	0	0	

from above table all entries in $x_3 \leq 0$

This shows that values in column x_3 increases infinitely.

∴ above Problem has unbounded soln.

III) Big - M Method :-

- Introduction:- Big - M Method is used to solve LPP that contain $\geq$ or $=$ type of constraints by adding artificial variables with a very large penalty (M), so they disappear from the final solⁿ.
- Notes:- 1) In case of problems either maximization type or minimization type, if there are constraints of the type ' $\geq$ ' or ' $=$ ', then introduce artificial variable.
2) In maximization problem the coefficient of each artificial variable taken as $-M$ where in minimization Problem the coefficient of each artificial variable is taken as a $+M$. Here, M is a very high positive No.
3) If in the solution, the artificial variable remains present in the basic variable, then given LPP has no solⁿ.

Que. 1) Solve the following LPP by using big-M Method.

$$\text{Min } Z = 60x_1 + 96x_2$$

$$\text{s.t. } 2x_1 + 4x_2 \geq 40$$

$$3x_1 + 3x_2 \geq 35$$

$$x_1, x_2 \geq 0.$$

Solⁿ:- Firstly, we write above LPP in standard form by adding Surplus and artificial variable.

$$\therefore \text{Min } Z = 60x_1 + 96x_2 + 0x_3 + 0x_4 + MR_1 + MR_2$$

s.t

$$2x_1 + 4x_2 - x_3 + 0x_4 + R_1 = 40$$

$$3x_1 + 3x_2 + 0x_3 - x_4 + R_2 = 35$$

$$x_1, x_2, x_3, x_4, R_1, R_2 \geq 0.$$

where, x_3 and x_4 are surplus variable and R_1 and R_2 are artificial variable.

Step: II) Initial Simplex Table.

CB	C_j	60	96	0	0	M	M	Soln	Ratio
	Basic Variable	x_1	x_2	x_3	x_4	R_1	R_2		
M	$\rightarrow R_1$	2	4	-1	0	1	0	40	10
M	R_2	3	3	0	-10	0	1	35	$35/3$
	Z_j	5M	4M	-M	-10M	M	M	45M	
	$C_j - Z_j$	60-5M	96-4M	M	M	-M	-M		

(opt $3R_1 - R_2 \rightarrow R_2, R_1 \rightarrow \frac{R_1}{4}$)

$\therefore$ all $C_j - Z_j \geq 0$ Not all $C_j - Z_j \geq 0$

$\therefore$ above soln is not optimal.

Thus, R_1 is leaving variable and x_2 is entering variable.

and 4 is a pivot element.

Step: III) 1st Simplex Table.

CB	C_j	60	96	0	0	M	M	Soln
	Basic Variable	x_1	x_2	x_3	x_4	R_1	R_2	
96	x_2	$\frac{1}{2}$	1	$-\frac{1}{4}$	0	$\frac{1}{4}$	0	10
M	$\rightarrow R_2$	$\frac{3}{2}$	0	$\frac{3}{4}$	-1	$-\frac{3}{4}$	1	5
	Z_j	$\frac{96}{2} + \frac{3}{2}M$	96	$\frac{3}{4}M - \frac{96}{4}$	-M	$\frac{96}{4} - \frac{3}{4}M$	M	$960 + 5M$
	$C_j - Z_j$	$12 - \frac{3}{2}M$	0	$24 - \frac{3}{2}M$	-M	$-24 + \frac{3}{2}M$	0	

$\therefore$ not all $C_j - Z_j \geq 0$, $\therefore$ The soln is not optimal.

$\therefore x_1$ is the entering variable.

and R_2 is leaving variable.

$\therefore \frac{3}{2}$ is a pivot element.

Step: IV) IIIrd Simplex Table.

CB	C _j	60	96	0	0	M	M	Soln
	Basic variable	x ₁	x ₂	x ₃	x ₄	R ₁	R ₂	
96	x ₂	0	1	-1/2	1/3	1/2	-1/3	25/3
60	x ₁	1	0	1/2	-2/3	-1/2	2/3	10/3
	Z _j	60	96	-18	-8	18	8	1000
	C _j -Z _j	0	0	18	8	M-18	M-8	

Here, all $\geq C_j - Z_j \geq 0$.

$\therefore$ This is an optimal soln.

$x_1 = \frac{10}{3}$, $x_2 = \frac{25}{3}$.

and $Z = 1000$.

Que. 2) Solve the following LPP by Simplex Big-M Method.

$$\text{Max } Z = 2x_1 + 4x_2$$

$$\text{s.t. } 2x_1 + x_2 \leq 18$$

$$3x_1 + 2x_2 \geq 30$$

$$x_1 + 2x_2 = 26$$

$$x_1, x_2 \geq 0$$

Soln:- Convert the LPP in standard form,

$$\text{Max } Z = 2x_1 + 4x_2$$

s.t.

$$2x_1 + x_2 + x_3 + 0x_4 + 0R_1 + 0R_2 = 18$$

$$3x_1 + 2x_2 + 0x_3 - x_4 + R_1 + 0R_2 = 30$$

$$x_1 + 2x_2 + 0x_3 + 0x_4 + 0R_1 + R_2 = 26$$

$$x_1, x_2, x_3, x_4, R_1, R_2 \geq 0$$

Step: II) Initial Simplex Table

CBi	Cj	2	4	0	0	-M	-M	Soln	Ratio
	XB	x_1	x_2	x_3	x_4	R_1	R_2		
0	x_3	2	1	1	0	0	0	18	18
-M	R_1	3	2	0	-1	1	0	30	15
-M	R_2	1	2	0	0	0	1	26	13
	Z_j	-4M	-4M	0	M	-M	-M	-56M	
	$C_j - Z_j$	2+4M	4+4M	0	-M	0	0		

Here, x_2 is entering variable, R_2 is leaving variable
and 2 is a pivot element.

Step: III) Ist Simplex Table

perform, $\frac{1}{2}R_3$, $R_1 - R_3$, $R_2 - 2R_3$.

CBi	Cj	2	4	0	0	-M	-M	Soln	Ratio
	XB	x_1	x_2	x_3	x_4	R_1	R_2		
0	x_3	$-\frac{3}{2}$	0	1	0	0	$-\frac{1}{2}$	5	—
-M	R_1	2	0	0	-1	1	-1	4	2
4	x_2	$\frac{1}{2}$	1	0	0	0	$\frac{1}{2}$	13	26
	Z_j	$-2M+2$	$4M$	0	$-M$	-M	$M+2$	$-4M+52$	
	$Z_j - C_j$	$2M$	0	0	-M	0	$-2M-2$		

Here, x_1 is the entering variable,
 R_1 is the leaving variable,
and 2 is a pivot element.

Step: IV) IInd Simplex Table

CB _i	C _j	2	4	0	0	0	0	-M	-M	
	X _B	x ₁	x ₂	x ₃	x ₄	R ₁	R ₂			Soln
0	x ₃	0	0	1	0	-9/2	3/2	-2		8
2	x ₁	1	0	0	-1/2	1/2	-1/2			2
4	x ₂	0	1	0	1/4	-1/4	3/4			12
	Z _j	2	4	0	0	0	2			52
	C _j -Z _j	0	0	0	0	M	2+M			

all $C_j - Z_j \geq 0$, $\therefore$ Current soln is optimum.

and soln is, $x_1 = 2$, $x_2 = 12$, $x_3 = 8$

and $Z = 52$.

Que. 3) Solve the following LPP.

$$\text{Min } Z = 3x_1 + 2x_2$$

$$\text{s.t. } 2x_1 + x_2 \leq 2$$

$$3x_1 + 4x_2 \geq 12$$

$$x_1, x_2 \geq 0$$

Soln:- Converting above problem into standard form,
we have,

$$\text{Min } Z = 3x_1 + 2x_2 + 0x_3 + 0x_4 - MR$$

s.t.

$$2x_1 + x_2 + x_3 + 0x_4 + 0R_1 = 2$$

$$3x_1 + 4x_2 + 0x_3 - x_4 + R_1 = 12$$

$$x_1, x_2, x_3, x_4, R_1 \geq 0$$

where, x_1 is slack variable,
 x_2 is surplus variable,
 R_1 is artificial variable.

$$\text{Min } Z =$$

Step: II) Initial Simplex Table

C_j		3	2	0	0	-M		
C_B	X_B	x_1	x_2	x_3	x_4	R_1	Soln	Ratio
0	x_3	2	1	1	0	0	2	2
-M	R_1	3	4	0	1	-M	12	3
	Z_j	$+3M$	$-4M$	0	M	-M	$-12M$	
	$C_j - Z_j$	$3+3M$	$4M+2$	0	-M	0		

Here, x_2 is entering variable,

x_3 is leaving variable.

1 is pivot element

Step: III) 1st Simplex Table

C_j		3	2	0	0	-M		
C_B	X_B	x_1	x_2	x_3	x_4	R_1	Soln	
2	x_2	2	1	1	0	0	2	
-M	R_1	-5	0	-4	-1	1	4	
	Z_j	$4+5M$	2	$4M+2$	M	-M	$-4M+2$	
	$Z_j - Z_j$	$-5M-1$	0	$-4M-2$	-M	0		

Here all $C_j - Z_j \leq 0$. $\therefore$ This is an optimal soln.

But the soln is,

$$x_2 = 2, R_1 = 4.$$

$\Rightarrow$ Artificial variable R_1 remains in the soln.

$\therefore$ In this case, we have no soln.

Duality :-

- Associated to every LPP there is another intimately related LPP is called the dual problem of the original LPP.
 - The original LPP is called the primal LPP.
- Note :-
- 1) The no. of constraint in one problem is equal to no. of variables in dual.
 - 2) The elements of the right hand side of the constraint in primal problem are the respective coefficients of the objective functions in the dual problem.
 - 3) If the primal problem is a minimization type, dual problem is maximization type.
 - 4) If the constraints in primal type are $\leq$ ($\geq$) then the constraints in the dual problem are of the type $\geq$ ($\leq$) except the feasibility constraints.
 - 5) The optimal solⁿ to a primal problem always provides the optimal solⁿ to its dual and vice versa.
- Remark :-
- 1) If objective function is of maximization type, then convert all constraints to ' $\leq$ ' type.
 - 2) If objective fn is of minimization type convert all constraints to ' $\geq$ ' type.
 - 3) Equality constraints i.e. of the type ' $a = b$ ' is equivalent to $a \leq b$ and $a \geq b$, write it as $a \leq b$ and $-a \leq -b$.

Que. 1) Write down the dual of following LPP.

1) Minimize $Z = 3x_1 + 25x_2$

s.t. $2x_1 + 4x_2 \geq 40$

$3x_1 + 2x_2 \geq 50$

$x_1, x_2 \geq 0$

Solⁿ. - The dual of given problem,

Maximize $Z^* = 40y_1 + 50y_2$

s.t.

$2y_1 + 3y_2 \leq 3$

$4y_1 + 2y_2 \leq 25$

$y_1, y_2 \geq 0$

2) Max $Z = 3x_1 + 4x_2$

s.t. $2x_1 + 3x_2 \leq 16$

$5x_1 + 2x_2 \geq 20$

$x_1, x_2 \geq 0$

Solⁿ. - Here, convert all constraints into ' $\leq$ ' type.

$\therefore$ Max $Z = 3x_1 + 4x_2$

s.t.

$2x_1 + 3x_2 \leq 16$

$-5x_1 - 2x_2 \leq -20$

$x_1, x_2 \geq 0$

Now, the dual of above problem is,

min $Z^* = 16y_1 - 20y_2$

s.t.

$2y_1 - 5y_2 \geq 3$

$3y_1 - 2y_2 \geq 4$

and $y_1, y_2 \geq 0$

3) Max $Z = x_1 - x_2 + 3x_3$

s.t. $-2x_1 + x_2 + 3x_3 = 2$

$-2x_1 + 3x_2 + 4x_3 = 1$

$x_1, x_2, x_3 \geq 0$

Soln:- firstly, convert all constraints into $\leq$ type

Above LPP can be written as,

Max $Z = x_1 - x_2 + 3x_3$

s.t.

$-2x_1 + x_2 + 3x_3 \geq 2$

$-2x_1 + x_2 + 3x_3 \leq 2$

$-2x_1 + 3x_2 + 4x_3 \geq 1$

$-2x_1 + 3x_2 + 4x_3 \leq 1$

Multiply, 1st and 3rd constraint by -1.

Max $Z = x_1 - x_2 + 3x_3$

s.t.

$2x_1 - x_2 - 3x_3 \leq -2$

$-2x_1 + x_2 + 3x_3 \leq 2$

$2x_1 - 3x_2 - 4x_3 \leq -1$

$-2x_1 + 3x_2 + 4x_3 \leq 1$

and, $x_1, x_2, x_3 \geq 0$.

Now, the dual of above LPP is

Min $Z^* = -2y_1 + 2y_2 - y_3 + y_4$

s.t.

$2y_1 - 2y_2 + 2y_3 - 2y_4 \geq 1$

$-y_1 + y_2 - 3y_3 + 3y_4 \leq -1$

$-3y_1 + 3y_2 - 4y_3 + 4y_4 \geq 3$

and, $y_1, y_2, y_3, y_4 \geq 0$.

IV) Two-Phase Method :-

Steps :-

Step: 0) Obtain the canonical form of the given Problem.
(Whatever may be the type of objective function, in the original Problem, only the minimization type of objective function is used in the two-phase method.)

Phase: I)

Step: I) Replace the objective fn with the sum of only artificial variables along with the same set of constraints of the canonical form of the original problem.

Step: II) Prepare the initial table for phase: I.

Step: III) Apply the usual simplex method till the optimality is reached.

Step: IV) Check whether the objective fn value is zero in the optimality table of phase I.

If **Yes** - Go to phase II.

If **No** - Conclude that the original problem has no feasible solⁿ and stop.

Phase: II)

Step: V) Drop the columns in the optimum table of phase I corresponding to the artificial variables which are currently non-basic.

Step: VI) If the some artificial variables are present at zero level in the basic solⁿ of the optimal table of phase I, substitute its objective fn coeff. with zero.

Step: VII) Substitute the coeff. of original objective fn in the optimal table of the phase I for the remaining variables and make necessary changes in CB column.

Examples:- Solve by using two-phase Method.

$$\min Z = 12x_1 + 18x_2 + 15x_3$$

$$\text{s.t.} \quad 4x_1 + 8x_2 + 6x_3 \geq 64$$

$$3x_1 + 6x_2 + 12x_3 \geq 96$$

$$x_1, x_2, x_3 \geq 0$$

Soln:- To write above LPP into standard form,

$$\min Z = 12x_1 + 18x_2 + 15x_3$$

$$\text{s.t.} \quad 4x_1 + 8x_2 + 6x_3 - x_4 + R_1 + 0x_5 + 0R_2 = 64$$

$$3x_1 + 6x_2 + 12x_3 + 0x_4 - x_5 + 0R_1 + R_2 = 96$$

$$x_1, x_2, x_3, x_4, x_5, R_1, R_2 \geq 0$$

Phase: I)

Step: I) Initial Simplex Table

	C_j	0	0	0	0	0	1	1	
C_B	X_B	x_1	x_2	x_3	x_4	x_5	R_1	R_2	
1	R_1	4	8	6	0	0	1	0	64
1	R_2	3	6	12	0	0	0	1	96
	Z_j	7	14	18	0	0	0	0	160
	$C_j - Z_j$	-7	-14	-18	0	0	0	0	

Step: II) 1st Simplex Table

	C_j	0	0	0	0	0	1	1	
C_B	X_B	x_1	x_2	x_3	x_4	x_5	R_1	R_2	
	R_1	5/2	5	0	-1	1/2	1	-1/2	16
	x_3	1/4	1/2	1	0	-1/12	0	1/12	8
	Z_j	5/2	5	0	-1	1/2	1	-1/2	16
	$C_j - Z_j$	-5/2	-5	0	1	-1/2	0	3/2	

Step: III) IInd Simplex Table

	C_j	0	0	0	0	0	1	1	
C_B	X_B	x_1	x_2	x_3	x_4	x_5	R_1	R_2	Sol ⁿ
0	x_2	$\frac{1}{2}$	1	0	$-\frac{1}{5}$	$\frac{1}{10}$	$\frac{1}{5}$	$-\frac{1}{10}$	$\frac{16}{5}$
0	x_3	0	0	1	$\frac{1}{10}$	$-\frac{2}{15}$	$-\frac{1}{10}$	$\frac{2}{15}$	$\frac{32}{15}$
	Z_j	0	0	0	0	0	0	0	0
	$C_j - Z_j$	0	0	0	0	0	ϕ	ϕ	

Here, set of basic variables in the optimal table of phase I does not contain artificial variable. so given problem has a feasible solⁿ.

Phase: II)

Step: IV) Initial Simplex Table for Phase-II

	C_j	12	18	15	0	0	
C_B	X_B	x_1	x_2	x_3	x_4	x_5	Sol ⁿ
18	x_2	$\frac{1}{2}$	1	0	$-\frac{1}{5}$	$\frac{1}{10}$	$\frac{16}{5}$
15	x_3	0	0	1	$\frac{1}{10}$	$-\frac{2}{15}$	$\frac{32}{15}$
	Z_j	9	18	15	$-\frac{21}{10}$	$-\frac{1}{5}$	$\frac{768}{5}$
	$C_j - Z_j$	3	0	0	$\frac{21}{10}$	$\frac{1}{5}$	

Transportation Problem :-

* Mathematical model for transportation Problem :-

It is presented as,

$$\text{Minimize } Z = \sum_{i=1}^m \sum_{j=1}^n C_{ij} x_{ij}$$

Generalized format of the transportation problem,

	Destination (j)
Source (i)	1 2 ... n
1	a ₁₁ a ₁₂ ... a _{1n}
2	a ₂₁ a ₂₂ ... a _{2n}
...	...
m	a _{m1} a _{m2} ... a _{mn}

Demand b₁ b₂ ... b_n

For Example :-

	Destination	Supply
Source	20 10 15	200
	10 12 9	300
	25 30 18	500
Demand	200 400 400	

* Terminology of Transportation Problem :-

1) Balanced Transportation Problem :-

A transportation problem is said to be balanced if the total supply from all the sources is equal to total demand in all destinations.

Ex: -

2) Unbalanced Transportation Problem :-

A transportation problem is said to be unbalanced if the total supply from all the sources is not equal to total demand in all destinations.

3) Feasible Solution :-

A set of all non-negative values x_{ij} , that satisfies the constraints is called as feasible soln.

4) Basic Feasible Solution (BFS) :-

A feasible soln of $m \times n$ transportation problem is said to be basic feasible soln if the total no. of positive allocations x_{ij} is exactly equal to $m+n-1$.

5) Optimal Solution :-

A feasible soln is said to be optimal if it minimize the total transportation cost.

6) An Initial Basic feasible Solution :-

A basic feasible soln that satisfies the supply and demand conditions is called as Initial basic feasible soln.

7) Degenerate Basic feasible solution :-

A basic feasible soln to a $m \times n$ transportation Problem is said to be Degenerate if no. of allocations are less than $(m+n-1)$.

8) Non - Degenerate Basic feasible soln:-

A basic feasible soln to a $m \times n$ transportation problem is said to be non-degenerate, if the no. of allocations are exactly $(m+n-1)$.

9) Dummy Source and Destination :-

Unbalanced TP can be converted into balanced TP by adding extra row/column with zero cost in each of its cell is called dummy source / destination.

* Methods for finding IBFS :-

There are three methods to find IBFS,

- 1) North West corner method (NWCN)
- 2) Matrix minima Method (Least cost Method)
- 3) Vogel's approximation method (VAM).

1) North West Corner Method (NWCN) :-

• Steps to solve :- 1) Consider a balanced TP if not then make it balanced by adding dummy row or column.

2) Consider the north-west corner cell, Allocate the minimum value of supply and demand to this cell.

3) After Allocations, check the following,

Case:1) If Supply of the row becomes zero, move downward to next row (Same column).

Case:2) If demand of the row becomes zero, move right to next column (Same row).

Case:3) If both supply and demand becomes zero, move diagonally to the next cell.

4) Repeat the same process until the demand and Supply are completely allocated.

Ex: 1) Obtain the IBFs for the following Problem by NWCM.

Source \ Destination	D ₁	D ₂	D ₃	D ₄	Supply
O ₁	2	3	11	7	6
O ₂	1	0	8	1	10
O ₃	5	8	15	9	17
Demand	7	5	3	2	

Soln: Given T.P. is balanced, as,

$$\text{Total Supply} = \text{Total Demand.}$$

$$\text{i.e. } \sum a_i = \sum b_j$$

Now, To Solve It By NWCM,

Source \ Destination	D ₁	D ₂	D ₃	D ₄	Supply
O ₁	2	3	11	7	6
O ₂	1	0	8	1	10
O ₃	5	8	15	9	17
Demand	7	5	3	2	

Repeat the same process until the demand and supply are completely allocated.

$$\begin{aligned}\text{Total minimum cost} &= (6 \times 2) + (1 \times 1) + (5 \times 8) + (3 \times 15) + (2 \times 9) + (6 \times 0) \\ &= 12 + 1 + 40 + 45 + 18 + 0 \\ &= 116 \text{ ₹. Soln} =\end{aligned}$$

2) Find the initial basic feasible solution by NWCM of the following TP.

Destination $\rightarrow$ origin $\downarrow$	D ₁	D ₂	D ₃	Supply
O ₁	13	15	16	29
O ₂	7	11	2	12
O ₃	19	20	9	16
Demand	14	8	23	

Soln:-

Here, Total Supply = $\sum b_i = 29 + 12 + 16$
 $= 45$

Total Demand = $\sum a_i = 14 + 8 + 23$
 $= 45$

$\therefore \sum a_i = \sum b_i$

$\therefore$ Given TP is balanced.

Destination Origin	D ₁	D ₂	D ₃	Supply
O ₁	13 (14)	15 (8)	16	29 14 8
O ₂	7	11 (5)	2 (7)	12 7
O ₃	19	20	9 (16)	16
Supply	14	8 5	23 16	

(0x3) $\therefore$ The total minimum cost $= (14 \times 13) + (3 \times 15) + (11 \times 5) + (7 \times 2)$
 $+ (16 \times 9)$
 $= 440 \text{ ₹}$

II) Matrix Minima Method :- (Least Cost Method)

Steps :- 1) In the cost matrix of T.P., find the cell with lowest transportation cost among all the rows or columns, (If the lowest cost is not unique, then the selection of the cell is arbitrary.)

2) Allocate as many units as possible to the selected cell.

- If supply is smaller $\rightarrow$ allocate full supply.
- If demand is smaller - allocate full demand.

3) If supply becomes 0 - Cancel (cross out) that row.

If demand becomes 0 - Cancel that column.

4) Repeat the process.

1) Find IBFs by least cost method of the following TP.

Source	Destination				supply
	D ₁	D ₂	D ₃	D ₄	
O ₁	2	3	11	7	6
O ₂	4	0	6	1	1
O ₃	5	8	15	9	10
Demand	7	5	3	2	

Soln:- Given TP is balanced.

Now, to find IBFS,

Source	Destination				Supply
	D ₁	D ₂	D ₃	D ₄	
O ₁	2 ⑥	3 -	11 -	4 -	6
O ₂	1 -	0 ①	6 -	1 -	8
O ₃	5 ①	8 ④	15 ③	9 ②	10
Demand	7	5	8	2	

Here, Total No. of allocations = $m+n-1$

$$= 4+3-1$$

$$= 6$$

∴ Given soln is IBFS.

Now, Total Transportation cost

$$= (2 \times 6) + (0 \times 1) + (5 \times 1) + (8 \times 4) + (15 \times 3) + (9 \times 2) \\ = 112 \text{ ₹}$$

3) Vogel's Approximation Method (VAM) :-

Steps :- 1) Check whether the problem is balanced.

If not, then make it balance by adding dummy row or dummy column.

2) Find Penalty for each row and column.

$$\text{Penalty} = \text{Second Smallest cost} - \text{Smallest cost.}$$

3) Select the highest Penalty.

4) Allocate in minimum cost cell.

5) Adjust supply and demand.

6) Repeat the process.

7) If the no. of allocations is ' $m+n-1$ ' then it is an IBFS, otherwise allocations are less than ' $m+n-1$ ', then solution is degenerate. We put ϵ in one of the unoccupied cells to remove degeneracy.

8) Calculate Transportation cost.

Que. 1) The following is an IBFS of a TP. check whether it is optimal. If not find optimal solution by MODI Method.

Warehouse $\rightarrow$	w_1	w_2	w_3	w_4	Supply
Factory $\downarrow$					
F_1	61	52	-2	3	70
F_2	2	4	0	1	38
F_3	1	2	-2	5	32
Demand	40	28	30	42	

Soln:-

Here, Total Supply = Total Demand

∴ This TP is balanced.

Warehouse → Origin ↓	w ₁	w ₂	w ₃	w ₄	Supply	Penalty
O ₁	1	2	-2	3	40	3 3
O ₂	2	4	0	1	38	1 1
O ₃	1	2	-2	5	32	3
Demand	40	28	30	42		
Penalty	0 1	0 2	0 2	2 2		

Warehouse	w ₁	w ₂	w ₃	w ₄	Supply	Penalty
O ₁	1 (38)	2 (28)	-2 (4)	3 (38)	40 38	3 1 1 1
O ₂	2	4	0 (38)	1	38	1 1 2 2
O ₃	1 (2)	2	-2 (30)	5	32 2	3 1 1 1
Demand	40 2	28	30 4	42 2		
Penalty	0 0 0	0 0 0	0 0 0	2 2 2		

$$\text{Total min cost} = (1 \times 38) + (28 \times 2) + (4 \times 3) + (1 \times 38) + (1 \times 2) + (30 \times (-2))$$

$$= 86$$

Optimum Solution :-

- An optimum Solution is obtained by making successive improvements in the IBFS, until no further decrease in transportation cost.

- Before testing the optimality, the IBFS must satisfy the following condⁿ -

- 1) The IBFS must contain ' $m+n-1$ ' positive allocations.

- 2) The allocations should be in the independent position; i.e. there is no closed loop through these allocations.

- closed loop means -

- a) Any two allocations should lie either in the same row or same column.

- b) Any three or more allocations should not lie in the same row or in the same column.

- c) Every loop has an even no. of allocations.

4) * MODI Method :-

Steps :- 1) Get IBFS and make it acceptable.

(Check above condⁿ)

- 2) For all allocated cells, find row and column value.

use, $C_{ij} = u_i + v_j$

where, C_{ij} = cost value of that allocated cell.

u_i = Row value

v_j = Column value.

Note:- Assume any one value = 0 (eg $v_1 = 0$).

- 3) Calculate Opportunity Cost :-

for every unallocated cell, calculate,

$$d_{ij} = C_{ij} - (u_i + v_j)$$

- 4) Check optimality :-

$$d_{ij} = C_{ij} - (u_i + v_j)$$

- If all $d_{ij} \geq 0$ $\rightarrow$ (solⁿ is optimal)

- If any $d_{ij} < 0 \rightarrow$ (solⁿ is not optimal)



Then select, most negative dij.

- 5) Start from the selected negative sign cell, form a closed loop using only allocated cell and marks sign alternately.
- 6) Find the minimum value in the cells marked with (-), subtract this value from (-) cells and add this value to (+) cells.
- 7) This gives a new allocation table, Repeat the process.

Que-1) The following is an IBFS of a TP. check whether it is optimal or not. If not find optimal soln by MODI Method.

Soln:-

Warehouse	w_1	w_2	w_3	w_4	Supply
f_1		6	5	8	5
f_2	5		11	9	7
f_3	8		9	7	13
Demand	35	28	32	25	

Soln:- Here, No. of allocations, $= m+n-1 = 6$

Also, allocations do not form a closed loop.
 $\therefore$ This is an IBFS.

Now, to check optimality,

Use equation, $C_{ij} = u_i + v_j$

Here, in 4th column, we get maximum allocations,
 $\therefore$ put $v_4 = 0$.

$\therefore$ we get, $u_1 = 5$, $u_2 = 7$, $u_3 = 13$.

$v_1 = -1$, $v_2 = 0$, $v_3 = -6$, $v_4 = 0$.

Now, we calculate opportunity cost for all unoccupied cells, by using the formula,

$$d_{ij} = C_{ij} - (u_i + v_j)$$

unoccupied cell

opportunity cost (d_{ij})

F_1W_1

$$6 - (5 - 2) = 3$$

F_1W_3

$$8 - (5 - 6) = 9$$

F_2W_2

$$11 - (7 - 0) = 4$$

F_2W_3

$$9 - (7 - 6) = 8$$

F_3W_1

$$8 - (13 - 2) = -3$$

F_3W_2

$$9 - (0 - 13) = -4$$

If all $d_{ij} \geq 0$, we get an optimal soln.

But, here the cell, F_2W_2 is the cell with largest negative opportunity cost among all unoccupied cell.

$\therefore$ we construct a closed loop with the cell F_3W_2 as follows,

Warehouse Factory	W_1	W_2	W_3	W_4	Supply	
F_1	6	5	8	5	30	$u_1 = 5$
F_2	5	11	9	7	40	$u_2 = 7$
F_3	8	9	17	13	50	$u_3 = 13$
Demand	35	28	32	25		

$v_1 = -2 \quad v_2 = 0 \quad v_3 = -6 \quad v_4 = 0$

on a closed loop, the minimum amount cell with negative sign is cell F_3W_4 , which has amount 18.

Now, we get next table as follows,

Warehouse	w_1	w_2	w_3	w_4	Supply
F_1	6	5	8	5	30
F_2	5	11	9	7	40
F_3	8	9	7	13	50
Demand	35	28	32	25	

Now, we test the solution for optimality,

find u_i, v_j , let $u_3 = 0$

$$u_1 = 4, u_2 = -2, u_3 = 0,$$

$$v_1 = 7, v_2 = 9, v_3 = 7, v_4 = 9.$$

Now, to calculate opportunity cost,

unoccupied cell	opportunity cost (d_{ij})
$F_1 w_1$	$6 - (4 - 7) = 3$
$F_2 w_3$	$8 - (7 - 4) = 5$
$F_2 w_2$	$11 - (9 - 2) = 4$
$F_2 w_3$	$9 - (7 - 2) = 4$
$F_3 w_1$	$8 - (7 + 0) = 1$
$F_3 w_4$	$13 - (9 + 0) = 4$

Here, all $d_{ij} \geq 0$.

$\therefore$ This is an optimum solution.

$$\therefore \text{Transportation cost} = (10 \times 5) + (20 \times 5) + (35 \times 5) + (5 \times 7) + (18 \times 9) + (32 \times 7)$$

$$= ₹465.$$